Asymptotics of Partition Functions of Coulomb Gases

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Dissertation's defence for the degree of Master of Sciences in Mathematics.
Supervisor: Guilherme Silva

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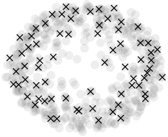
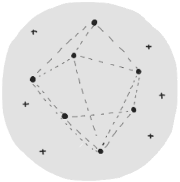
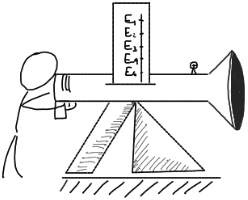


Random Matrices

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E_2
E_3
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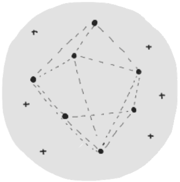
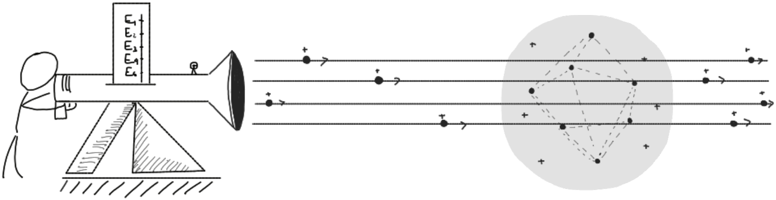
E_5

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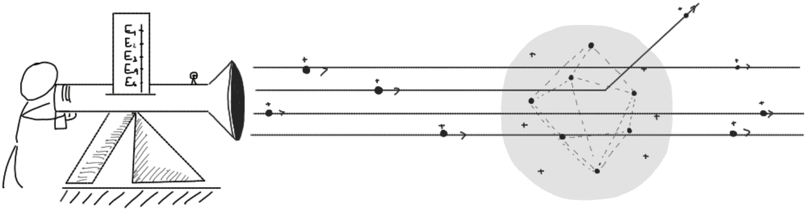

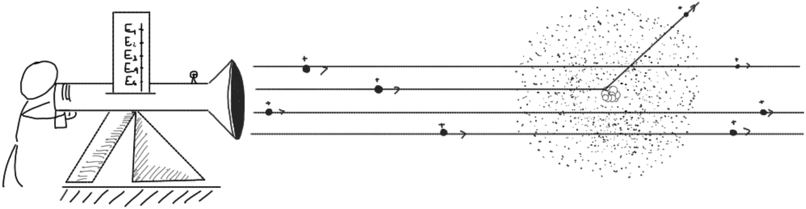
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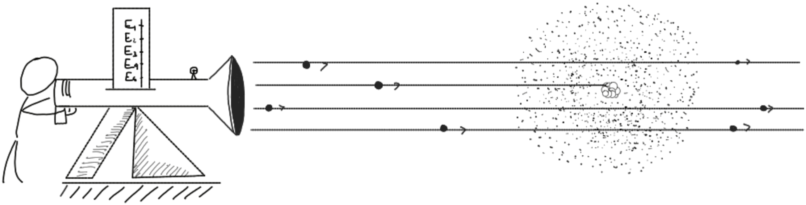
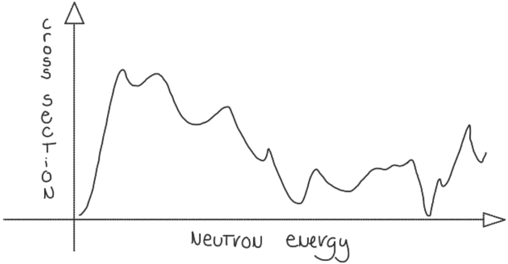
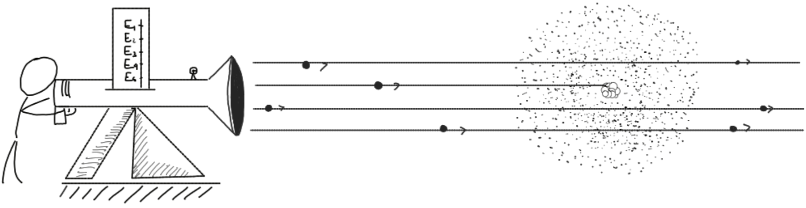
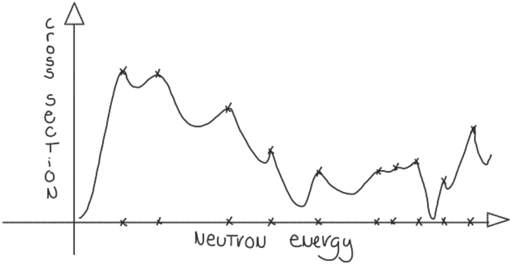



cross section
neutron energy

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Eugene Wigner - *On the Wigner Surmise* (1956)

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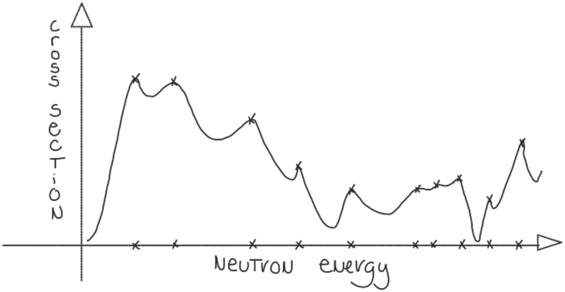

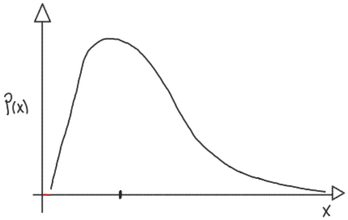

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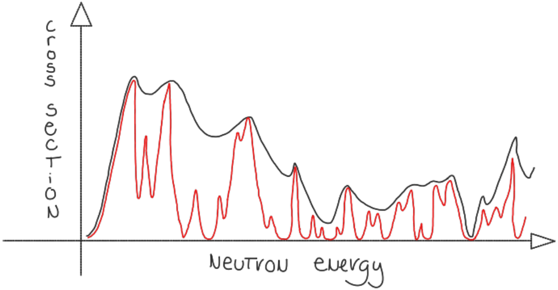

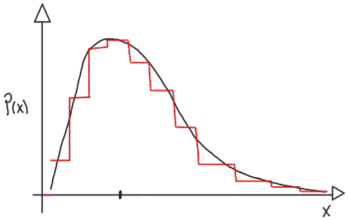

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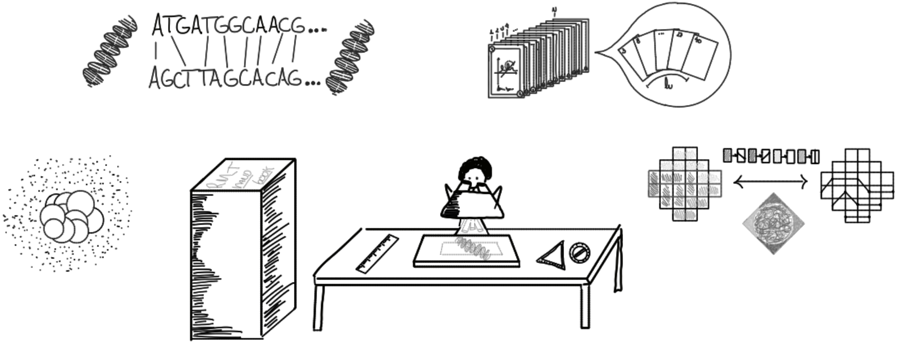

A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} & x_{1,4} \\ x_{2,1} & x_{2,2} & x_{2,3} & x_{2,4} \\ x_{3,1} & x_{3,2} & x_{3,3} & x_{3,4} \\ x_{4,1} & x_{4,2} & x_{4,3} & x_{4,4} \\ x_{5,1} & x_{5,2} & x_{5,3} & x_{5,4} \end{bmatrix}$$

$\mathrm{p}(x_{i,j})$	$\mathbb{S}$	$\mathcal{M}_n$
Discrete	$\mathbb{R}$	General
Uniform	$\mathbb{C}$	Hermitian
Gaussian	$\mathbb{H}$	Normal
$\vdots$	$\vdots$	$\vdots$

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A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} -1 & -1 & 1 & 1 \\ -1 & 1 & -1 & 1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$$

$\mathrm{p}(x_{i,j})$	$\mathbb{S}$	$\mathcal{M}_n$
Discrete	$\mathbb{R}$	General
Uniform	$\mathbb{C}$	Hermitian
Gaussian	$\mathbb{H}$	Normal
⋮	⋮	⋮

$$\mathcal{P}(\hat{X}) = \prod_{i,j=1}^{n} p(x_{i,j}) = U^{n^2}(\{0,1\})$$

A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} 0.213 & 0.368 & 0.912 & 0.881 \\ 0.759 & 0.692 & 0.000 & 0.332 \\ 0.898 & 0.987 & 0.422 & 0.623 \\ 0.145 & 0.880 & 0.522 & 0.686 \\ 0.753 & 0.687 & 0.787 & 0.515 \end{bmatrix}$$

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$$\begin{bmatrix} -0.022 & -1.340 & -0.483 & -1.134 \\ 1.752 & -1.644 & 1.081 & -0.936 \\ -1.058 & -0.091 & 0.544 & -0.382 \\ 0.633 & 0.567 & 0.606 & -0.621 \\ -0.316 & -1.128 & -0.150 & 0.251 \end{bmatrix}$$

$\mathrm{p}(x_{i,j})$	$\mathbb{S}$	$\mathcal{M}_n$
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Ginibre Orthogonal Ensemble

$$\mathcal{P}(\hat{X}) = \prod_{i,j=1}^{n} \mathrm{e}^{-nx_{i,j}^2}$$

A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} -2.0+0.1i & -0.2-0.9i & -0.4-0.3i & -0.3+1.0i \\ -1.6-0.8i & 0.3-0.6i & -1.7+0.9i & -0.1+0.3i \\ -0.5-0.9i & -0.5+0.2i & 1.2+1.0i & 0.4-1.3i \\ -0.6+2.4i & 0.1-0.7i & -0.2+1.6i & 0.6-0.4i \\ 0.1-1.3i & -0.2-0.9i & 1.2-1.6i & 2.1+2.0i \end{bmatrix}$$

$\mathrm{p}(x_{i,j})$	$\mathbb{S}$	$\mathcal{M}_n$
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⋮	⋮	⋮

Ginibre Unitary Ensemble

$$\mathcal{P}(\hat{X}) = \prod_{i,j=1}^{n} \mathrm{e}^{-nx_{i,j}^2}\, \mathrm{e}^{-ny_{i,j}^2}$$

A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} -0.8 & -0.2-1.5i & 0.9-0.6i & -3.2+2.3i \\ -0.2+1.5i & 0.5 & -1.8-0.8i & 1.5+1.8i \\ 0.9+0.6i & -1.8+0.8i & 1.8 & 2.1+2.5i \\ -3.2-2.3i & 1.5-1.8i & 2.1-2.5i & 2.4 \end{bmatrix}$$

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Gaussian Unitary Ensemble

$$\mathcal{P}(\hat{X}) = \prod_{i=1}^{n} \mathrm{e}^{-nz_{i,j}^2} \prod_{i>j}^{n} \mathrm{e}^{-2nz_{i,j}^2}$$

We can think of the matrix itself as a random variable, a measurable map from a probability measure space to $\mathcal{M}_n$.

$$\begin{bmatrix} -0.8 & -0.2-1.5i & 0.9-0.6i & -3.2+2.3i \\ -0.2+1.5i & 0.5 & -1.8-0.8i & 1.5+1.8i \\ 0.9+0.6i & -1.8+0.8i & 1.8 & 2.1+2.5i \\ -3.2-2.3i & 1.5-1.8i & 2.1-2.5i & 2.4 \end{bmatrix}$$

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$e^{-n\,\mathrm{tr}\left\{(\hat{X}^*\hat{X})\right\}}$	$\mathbb{C}$	Hermitian
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Gaussian Unitary Ensemble

$$\mathcal{P}(\hat{X}) = \prod_{i=1}^{n} e^{-nz_{i,j}^2} \prod_{i>j}^{n} e^{-2nz_{i,j}^2} \propto e^{-n\,\mathrm{tr}\left(\hat{X}^2\right)}$$

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We can think of the matrix itself as a random variable, a measurable map from a probability measure space to $\mathcal{M}_n$.

Normal matrices are similar to diagonal matrices, i.e.

$$\hat{X} = \hat{U}\hat{\Lambda}\hat{U}^*.$$

Moreover, the spectra is typically simple.

$\mathcal{P}(\hat{X})$	$\mathbb{S}$	$\mathcal{M}_n$
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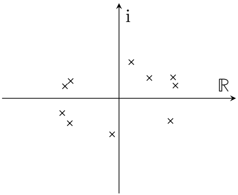
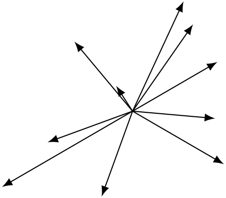
$$\hat{X} \mapsto (\hat{\Lambda}, \hat{U}) = \left(\quad , \quad \right)$$





We can think of the matrix itself as a random variable, a measurable map from a probability measure space to $\mathcal{M}_n$.

Invariance by unitary conjugation means independence of coordinates.

$$\mathrm{d}\mathcal{P}(\hat{X}_1) = \mathrm{d}\mathcal{P}(\hat{X}_2) \iff \hat{X}_1 = \hat{U}\hat{X}_2\hat{U}^*$$

for some unitary U.

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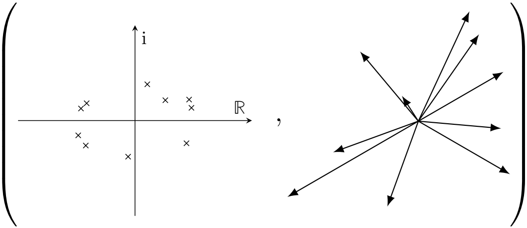
$\hat{X} \mapsto (\hat{\Lambda}, \hat{U}) =$ 

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$$\begin{aligned}\exp\left(-n\,\mathrm{tr}\left\{Q(\hat{U}\hat{X}_1\hat{U}^*)\right\}\right) &= \exp\left(-n\,\mathrm{tr}\left\{Q(\hat{U}^*\hat{U}\hat{X}_1)\right\}\right) = \exp\left(-n\,\mathrm{tr}\left\{Q(\hat{X}_1)\right\}\right) \\ &= \exp\left(-n\,\mathrm{tr}\left\{Q(\hat{U}\hat{\Lambda}\hat{U}^*)\right\}\right) = \exp\left(-n\sum_i^n Q(\lambda_i)\right).\end{aligned}$$

We can think of the matrix itself as a random variable, a measurable map from a probability measure space to $\mathcal{M}_n$.

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We want that $\mathrm{d}\mathcal{P}(\hat{X}) = \mathcal{P}(\hat{X})\mathrm{d}\hat{X}$ is invariant.

$$\mathrm{d}\hat{X} = \prod_{1\le i<j\le n} |z_i - z_j|^2 \mathrm{d}\hat{U} \prod_{i=1}^{n} \mathrm{d}^2 z_i,$$

where $\mathrm{d}\hat{U}$ is the Haar measure on eigenvectors and $\mathrm{d}^2 z$ is Lebesgue.

We set the ensemble

$$\mathcal{E} = (\mathcal{S}, \mathcal{P}) \quad \text{with} \quad \begin{cases} \mathcal{S} = \mathcal{N}_n(\mathbb{C}) \\ \mathcal{P}(\hat{X}) = \exp\left\{-n \operatorname{tr}\Big(Q(\hat{X})\Big)\right\} \end{cases}$$

such that

$$\mathrm{d}\mathcal{P}(\hat{X}) = \prod_{i<j}^{n} |z_i - z_j|^2 \prod_{j=1}^{n} \mathrm{e}^{-nQ(z_j)}\, \mathrm{d}^2 z_j \mathrm{d}\hat{U}.$$

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$$\begin{aligned} \int_{\mathcal{M}_n} \mathrm{d}\mathcal{P}(\hat{X}) &= \int_{\mathbb{C}^n} \int_{\tilde{U}(n)} \prod_{i<j}^{n} |z_i - z_j|^2 \prod_{j=1}^{n} \mathrm{e}^{-nQ(z_j)} \, \mathrm{d}^2 z_j \mathrm{d}\hat{U} \\ &= \int_{\mathbb{C}^n} \frac{\prod_{i<j}^{n} |z_i - z_j|^2 \prod_{j=1}^{n} \mathrm{e}^{-nQ(z_j)}}{\mathcal{Z}_n^Q} \mathrm{d}^2 z_j \end{aligned}$$

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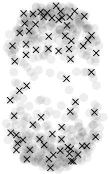
we have

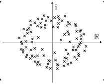
$$p_n(\{z_i\}_{i=1}^n) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i<j}^{n} |z_i - z_j|^2 \prod_{j=1}^{n} \mathrm{e}^{-nQ(z_j)}$$



Modeling our Model

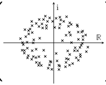
We know that

$$p_n(\{z_i\}_{i=1}^n) = p\Bigg($$



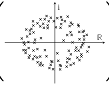
$$\Bigg) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i<j}^n |z_i - z_j|^2 \prod_{j=1}^n \mathrm{e}^{-nQ(z_j)}$$

We know that

$p_n(\{z_i\}_{i=1}^n) = p\Bigg($ 

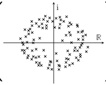
$\Bigg) = \dfrac{1}{\mathcal{Z}_n^Q} \prod_{i<j}^{n} |z_i - z_j|^2 \prod_{j=1}^{n} \mathrm{e}^{-nQ(z_j)}$

We can notice two effects, one coming from the Jacobian and other from Q.

$p_n\Bigg($ 

$\Bigg) > \Bigg\{$

We know that

$p_n(\{z_i\}_{i=1}^n) = p\Bigg($ 

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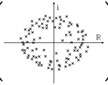
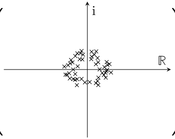
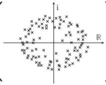
$p_n\Bigg($ 

$\Bigg) > \Bigg\{ \; p_n\Bigg($ 

$\Bigg),$ the interactions dominates.

We know that

$p_n(\{z_i\}_{i=1}^n) = p\Bigg($ 

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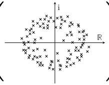
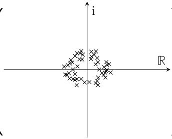
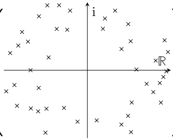
$p_n\Bigg($ 

$\Bigg) > \Bigg\{$

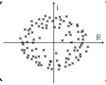
$p_n\Bigg($ 

$\Bigg)$, the interactions dominates.

$p_n\Bigg($ 

$\Bigg)$, the external potential dominates.*

We know that

$p_n(\{z_i\}_{i=1}^n) = p\Bigg($ 

$\Bigg) = \dfrac{1}{\mathcal{Z}_n^Q} \prod_{i<j}^{n} |z_i - z_j|^2 \prod_{j=1}^{n} \mathrm{e}^{-nQ(z_j)}$

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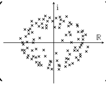

$$\Bigg) = \frac{1}{\mathcal{Z}_n^Q}\prod_{i<j}^{n}|z_i - z_j|^2\prod_{j=1}^{n}\mathrm{e}^{-nQ(z_j)}$$

$$= \frac{1}{\mathcal{Z}_n^Q}\exp\left\{-n^2\left(\frac{1}{n}\sum_{i=1}^{n}Q(z_i) + \frac{1}{n^2}\sum_{i\neq j}\ln|z_i - z_j|\right)\right\}$$

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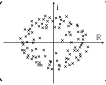

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$$= \frac{\mathrm{e}^{-n^2 \mathcal{H}(\chi_n)}}{\mathcal{Z}_n^Q}, \quad \mathcal{H}(\chi_n) = \frac{1}{n} \sum_{i=1}^{n} Q(z_i) + \frac{1}{n^2} \sum_{i\neq j}^{n} \ln|z_i - z_j|$$

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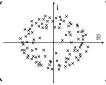
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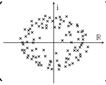
The measure $\mathrm{d}p_n$ models an Coulomb-interacting gas of charged indistinguishable particles under some external field Q on $\mathbb{C}$.

We know that

$$p_n(\{z_i\}_{i=1}^n) = p\Bigg($$



$$\Bigg) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i<j}^{n} |z_i - z_j|^2 \prod_{j=1}^{n} \mathrm{e}^{-nQ(z_j)}$$

$$= \frac{1}{\mathcal{Z}_n^Q} \exp\left\{ -n^2 \left(\frac{1}{n} \sum_{i=1}^{n} Q(z_i) + \frac{1}{n^2} \sum_{i \neq j} \ln |z_i - z_j| \right) \right\}$$

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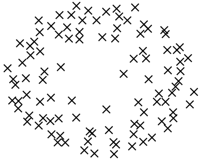
The measure $\mathrm{d}p_n$ models an Coulomb-interacting gas of charged indistinguishable particles under some external field Q on $\mathbb{C}$.

If we want to study this gases, how to describe its statistical properties?

Point Process

Suppose there are n particles at random locations in the complex plane $\mathbb{C}$ denoted by the configuration $\chi_n = \{z_1, z_2, \ldots, z_n\}$.



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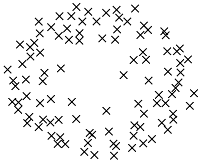

Define a quantity $\xi(A)$ s.t.

$$\xi(A) = \sum_{k=1}^{n} \delta(z_k)$$

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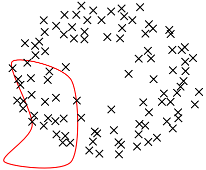
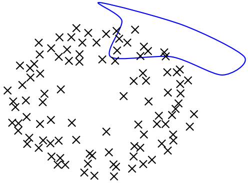


Define a quantity $\xi(A_1)$ s.t.

$$\xi(A_1) = \sum_{k=1}^{n} \delta_{A_1}(z_k) = 21$$

Point Process

Suppose there are n particles at random locations in the complex plane $\mathbb{C}$ denoted by the configuration $\chi_n = \{z_1, z_2, \ldots, z_n\}$.

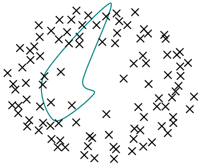


Define a quantity $\xi(A_2)$ s.t.

$$\xi(A_2) = \sum_{k=1}^{n} \delta_{A_2}(z_k) = 9$$

Point Process

Suppose there are n particles at random locations in the complex plane $\mathbb{C}$ denoted by the configuration $\chi_n = \{z_1, z_2, \ldots, z_n\}$.

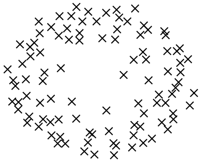


Define a quantity $\xi(A_3)$ s.t.

$$\xi(A_3) = \sum_{k=1}^{n} \delta_{A_3}(z_k) = 10$$

Point Process

Suppose there are n particles at random locations in the complex plane $\mathbb{C}$ denoted by the configuration $\chi_n = \{z_1, z_2, \ldots, z_n\}$.



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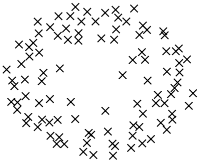
$$\xi(A) = \sum_{k=1}^{n} \delta(z_k)$$

This defines a *random counting measure* in the usual sense.

Point Process

Suppose there are n particles at random locations in the complex plane $\mathbb{C}$ denoted by the configuration $\chi_n = \{z_1, z_2, \dots, z_n\}$.

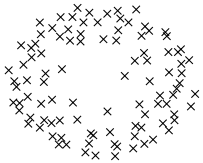


Define a quantity $\xi(A)$ s.t.

$$\xi(A) = \sum_{k=1}^{n} \delta(z_k)$$

This defines a *random counting measure* in the usual sense. A **point process** $\mathbb{P}$ on $\mathbb{C}$ is a probability measure on the space of all counting measures $\mathcal{N}(\mathbb{C})$.

Consider the configuration $\chi_n = \{z_1, z_2, \ldots, z_n\}$.



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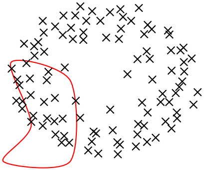
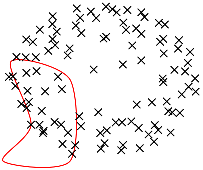


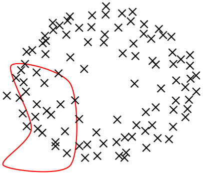
$$\xi(A_1) = \sum_{k=1}^{n} \delta_{A_1}(z_k)$$

Consider the configuration $\chi_n^{(1)} = \{z_1^{(1)}, z_2^{(1)}, \dots, z_n^{(1)}\}$.



$$\xi^{(1)}(A_1) = \sum_{k=1}^{n} \delta_{A_1}(z_k^{(1)})$$

Consider the configuration $\chi_n^{(2)} = \{z_1^{(2)}, z_2^{(2)}, \dots, z_n^{(2)}\}$.



$$\xi^{(2)}(A_1) = \sum_{k=1}^{n} \delta_{A_1}(z_k^{(2)})$$

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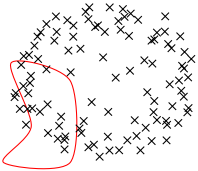

$$\xi^{(3)}(A_1) = \sum_{k=1}^{n} \delta_{A_1}(z_k^{(3)})$$

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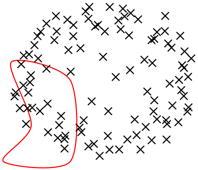
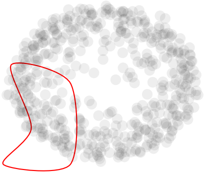


$$\xi(A_1) = \sum_{k=1}^{n} \delta_{A_1}(z_k)$$

What is

$$\mathfrak{p}(A) = \mathbb{E}[\xi(A)] := \mathbb{E}[\#(\chi_n \cap A)], \quad A \in \mathcal{E}.$$

Consider the configuration $\chi_n = \{z_1, z_2, \dots, z_n\}$.



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We can assume the existence of a function $\mathrm{p}_1(z)$ s.t.

$$\mathfrak{p}(A) = \int_A \mathrm{p}_1(z) \mathrm{d}^2 z$$

where $\mathrm{d}^2 z$ is the usual area measure.

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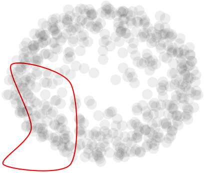

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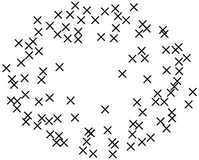
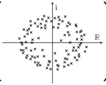
$$\mathfrak{p}(A) = \int_A \mathrm{p}_1(z) \mathrm{d}^2 z$$

where $\mathrm{d}^2 z$ is the usual area measure. The 1-**point correlation function** p_1 represents the infinitesimal probability of a point of the process being at a location z, i.e. $\mathbb{P}(\xi(\mathrm{d}^2 z) = 1) = \mathrm{p}_1(z)\mathrm{d}^2 z$.

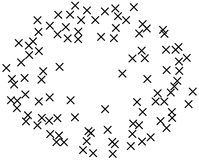
Variational Problem

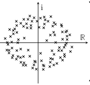


$p\Bigg($  $\Bigg) = \dfrac{e^{-n^2\mathcal{H}(\chi_n)}}{\mathcal{Z}_n^Q},$

$$\mathcal{H}(\chi_n) = \frac{1}{n^2}\sum_{i\neq j}^{n} \ln|z_i - z_j| + \frac{1}{n}\sum_{i=1}^{n} Q(z_i).$$

Variational Problem



$p\Bigg($ 

$\Bigg) = \dfrac{e^{-n^2\mathcal{H}(\chi_n)}}{\mathcal{Z}_n^Q},$

$$\frac{1}{n}\sum_{k=1}^{n}\delta(z_k) = \mu_n$$

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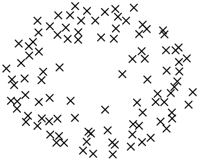
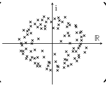
Variational Problem



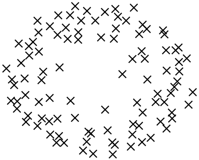
$p\Bigg($ 

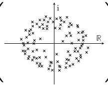
$\Bigg) = \dfrac{e^{-n^2\mathcal{H}(\chi_n)}}{\mathcal{Z}_n^Q},$

$$\frac{1}{n}\sum_{k=1}^{n}\delta(z_k) = \mu_n$$

$$\mathcal{H}(\chi_n) = \int\int_{t\neq s} \ln|t-s|^{-1}\mathrm{d}\mu_n(t)\mathrm{d}\mu_n(s) + \int Q(s)\mathrm{d}\mu_n(s).$$

Variational Problem





$$p\left(\quad \right) = \frac{e^{-n^2\mathcal{H}(\chi_n)}}{\mathcal{Z}_n^Q},$$

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} \delta(z_k) = \lim_{n\to\infty} \mu_n = \mu$$

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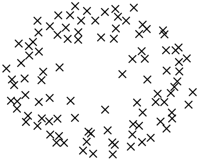
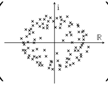
Variational Problem



$p\left(\right.$  $\left.\right) = \dfrac{e^{-n^2\mathcal{H}(\chi_n)}}{\mathcal{Z}_n^Q}$,

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} \delta(z_k) = \lim_{n\to\infty} \mu_n = \mu$$

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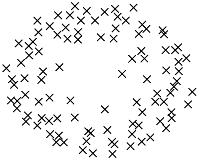
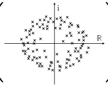
Variational Problem



$$p\left(\right.$$  $$\left. \right) = \frac{e^{-n^2\mathcal{H}(\chi_n)}}{\mathcal{Z}_n^Q},$$

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} \delta(z_k) = \lim_{n\to\infty} \mu_n = \mu$$

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Associated with the variational problem defined by

$$E^Q = \inf_{\mu\in\mathcal{M}^1}\left[\int\int_{t\neq s} \ln|t-s|^{-1}\mathrm{d}\mu(t)\mathrm{d}\mu(s) + \int Q(s)\mathrm{d}\mu(s)\right] = I_Q(\mu_Q),$$

one defines the so-called *Euler-Lagrange* type variational equations.

Theorem

(Strong Variational Equations) Suppose that $\mathrm{d}\mu_Q = \mu(z)\mathrm{d}^2 z$ for some continuous function μ of compact support. The equilibrium measure μ_Q for the given variational problem, where $\Sigma = \mathbb{C}$ satisfies the following conditions. There exists a real constant l such that

1. $$2\int \ln|z-t|^{-1}\mathrm{d}\mu_Q(t) + Q(z) \geq l$$

   for all $z \in \mathbb{C}$.

2. $$2\int \ln|z-t|^{-1}\mathrm{d}\mu_Q(t) + Q(z) = l$$

   on $\{z : \mu(z) > 0\}$.

We define $U^\mu(z) = \int \ln|z-t|^{-1}\mathrm{d}\mu_Q(t)$.

Normal Ensemble - Rotational symmetry

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3. Furthermore, assume Q to be smooth, sub-harmonic in $\mathbb{C}$, and strictly sub-harmonic in a neighborhood of the $\mathrm{supp}(\mu_Q)$.

Corollary

If r_0 is the largest solution of $rq'(r) = 0$ and r_1 is the smallest solution of $rq'(r) = 2$ then the support of μ_Q is given by the ring $\mathcal{S} = \{z | r_0 \leq |z| \leq r_1\}$ and $\mathrm{d}\mu_Q$ is given by

$$\mathrm{d}\mu_Q(z) = \frac{1}{4\pi}(rq'(r))'\delta_{\mathcal{S}}\mathrm{d}r\mathrm{d}\phi, \quad z = r\,\mathrm{e}^{\mathrm{i}\phi}\,.$$

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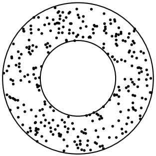

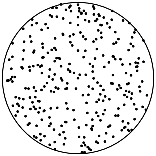

We call **Droplet** the support of μ_Q, which in our case is the set of all complex values z such that $r_0 \leq |z| \leq r_1$.

Let $\mu \in \mathcal{M}^1(\mathbb{C})$ be a measure such that $\mathrm{d}\mu = \frac{1}{4\pi}(rq'(r))'\delta_{\mathcal{S}}\mathrm{d}\phi\mathrm{d}r$.

$$\int_{\mathbb{C}} \mathrm{d}\mu = 1,$$

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$$\begin{aligned} 2U^{\mu}(z) &\geq l - Q(z), \quad z \in \mathbb{C} \\ 2U^{\mu}(z) &= l - Q(z), \quad z \in \{z \mid r_0 \leq |z| \leq r_1\}. \end{aligned}$$

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with $I_Q(\mu) < \infty$, then $\mu_Q = \mu$. We must compute

$$U^{\mu}(z) = \int \ln |z - t|^{-1} \mathrm{d}\mu_Q(t) = \frac{1}{4\pi} \int_{r_0}^{r_1} (rq'(r))' \int_{-\pi}^{\pi} \ln \frac{1}{|z - r\,\mathrm{e}^{\mathrm{i}\phi}|} \mathrm{d}\phi \mathrm{d}r.$$

With the formula

$$\int_{-\pi}^{\pi} \ln \frac{1}{|z - r\,e^{i\phi}|} d\phi = \begin{cases} 2\pi \ln \frac{1}{r}, & |z| \leq r, \\ 2\pi \ln \frac{1}{|z|}, & |z| > r. \end{cases}$$

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$$\begin{aligned} U^{\mu}(z) &= \frac{1}{2}\int_{r_0}^{|z|} (rq'(r))' \ln\left(\frac{1}{r}\right)\mathrm{d}r + \frac{1}{2}\int_{|z|}^{r_1} (rq'(r))' \ln\left(\frac{1}{z}\right)\mathrm{d}r, \\ &= -\frac{1}{2}(2\ln(r_1) - q(r_1)) - \frac{1}{2}q(|z|). \end{aligned}$$

Define $l = -(2\ln(r_1) - q(r_1))$, then

$$2U^\mu(z) = \begin{cases} l - q(r_0), & |z| < r_0, \\ 2\ln\frac{1}{|z|}, & |z| > r_1, \\ l - q(|z|), & r_0 \le |z| \le r_1. \end{cases}$$

We need to prove that for $z \in \mathbb{C}$, we have $2U^\mu(z) \ge l - q(|z|)$ while on $r_0 \le |z| \le r_1$ we have $2U^\mu(z) = l - q(|z|)$.

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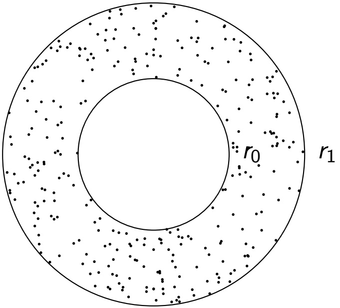

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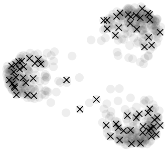
$$q(|z|) - 2\log(|z|) \geq q(r_1) - 2\log(r_1)$$
$$\implies 2\ln\frac{1}{|z|} \geq l - q(|z|)$$



Asymptotics on the Partition Function

Quick recap

$$\mathrm{d}\mathcal{P}(\hat{X}) = \prod_{i<j}^{n} |z_i - z_j|^2 \prod_{j=i}^{n} \mathrm{e}^{-nQ(z_i)} \, \mathrm{d}^2 z_i \mathrm{d}\hat{U}.$$

$\mathcal{P}(\hat{X})$	$\mathbb{S}$	$\mathcal{M}_n$
$\mathrm{e}^{-n\,\mathrm{tr}\left(\hat{M}^2\right)}$	$\mathbb{R}$	General
$\mathrm{e}^{-n\,\mathrm{tr}\left\{(\hat{X}^*\hat{X})\right\}}$	$\mathbb{C}$	Hermitian
$\mathrm{e}^{-n\,\mathrm{tr}\left\{(Q(\hat{X}))\right\}}$	$\mathbb{H}$	Normal
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Orthogonal Polynomial Ensemble

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$$p_n(\{z_i\}_{i=1}^n) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i<j}^n |z_i - z_j|^2 \prod_{j=i}^n e^{-nQ(z_i)}$$

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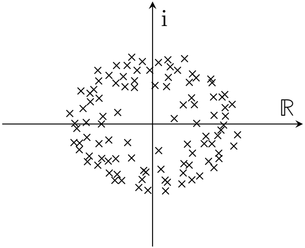

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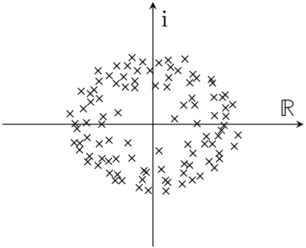

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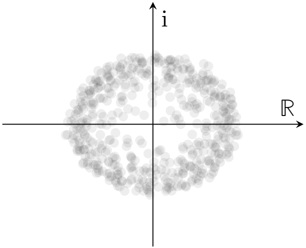

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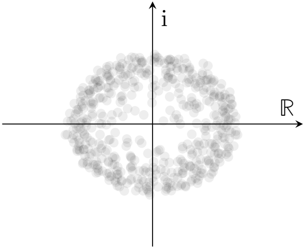

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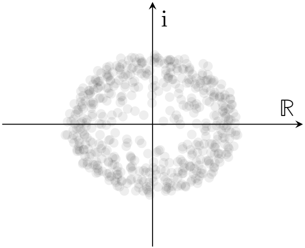

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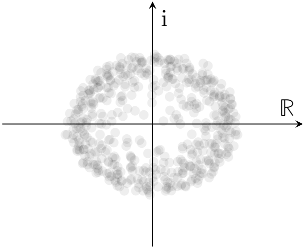

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$$\ln\left(\mathcal{Z}_n^Q\right) = -I_Q n^2 + \frac{1}{2} n \ln n + E_Q n + G \ln n + F_Q + o(1).$$

Sketch

1. *Partition functions can be expressed in terms of the norming constants*: We can write the free energy as

$$\ln \mathcal{Z}_n^Q = \ln n! + \sum_{j=0}^{n-1} \ln h_j.$$

2. *Asymptotics of norming constant using the asymptotic Laplace technique* .

$$h_j = \int_{\mathbb{C}} |z|^{2j} \, \mathrm{e}^{-nQ(z)} \, \mathrm{d}^2 z \approx \frac{\mathrm{e}^{-nV_{\tau(j)}(r_c(\tau(j)))}}{\sqrt{n}}.$$

3. *Asymptotic of the summation on* $\ln h_j$ *using quadrature techniques*:

$$-\sum_{j=0}^{n-1} \ln h_j \approx -n^2 \int_{r_0}^{r_1} V(r) \frac{(rq'(r))'}{2} \mathrm{d}r = -n^2 I_Q[\mu_Q].$$

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Remember that the j.p.d.f. considered, with $w(z_i) = e^{-nQ(z_i)}$, is

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$$p_n(z_1, z_2, \ldots, z_n) = \frac{\prod_{l=1}^{n} h_{n,l}}{\mathcal{Z}_n^Q} \det\left[K(z_i, z_j)\right]_{n\times n}$$

With $\prod_{i<j}^{n} |z_i - z_j| = \det\left(q_{j-1}(z_i)\right)_{n\times n}$ and $(\det(A))^2 = \det\Big(\sum_{j=1}^{n} \overline{A_{ji}} A_{jk}\Big)$,

$$\frac{w(z_i)}{\prod_{l=1}^{n} h_{n,l}} \prod_{i<j}^{n} |z_i - z_j|^2 = \det\left(K(z_i, z_j)\right)_{n\times n}$$

where q_i form a polynomial basis for the space.

Determinantal P.P.

Remember that the j.p.d.f. considered, with $w(z_i) = \mathrm{e}^{-nQ(z_i)}$, is

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Theorem

Let $\{q_{j,n}\}_{j=0}^{\infty}$ be a family of monic polynomials such that each $q_j(\lambda)$ is of degree j. Impose also that exists finite positive constants $\{h_{j,n}\}_{j=0}^{\infty}$ s.t.

$$\langle q_{j,n}, q_{k,n}\rangle_w = \int_{\mathbb{C}} q_{j,n}(z)\overline{q_{k,n}(z)}\,\mathrm{e}^{-nQ(z)}\,\mathrm{d}^2 z = h_{j,n}\delta_{\{j=k\}}$$

for any $j, k > 0$. Then, $K_n(x, y)$ has reproducing properties, i.e.

$$\int_{\mathbb{C}} \det\left[K(\lambda_i, \lambda_j)\right]_{i,j=1}^{k+1} \mathrm{d}^2\lambda_{k+1} = (n-k)\det\left[K(\lambda_i, \lambda_j)\right]_{i,j=1}^{k}.$$

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Corollary

Because of

$$\int_{\mathbb{C}} \det\left[K(\lambda_i, \lambda_j)\right]_{i,j=1}^{k+1} \mathrm{d}^2\lambda_{k+1} = (n-k) \det\left[K(\lambda_i, \lambda_j)\right]_{i,j=1}^{k},$$

the partition function $\mathcal{Z}_n^Q$ can be written as

$$\mathcal{Z}_n^Q = \prod_{l=1}^{n} h_{n,l} \int_{\mathbb{C}^n} \det\left[K(\lambda_i, \lambda_j)\right]_{i,j=1}^{n} \prod_{i=1}^{n} \mathrm{d}^2\lambda_i = n! \prod_{l=1}^{n} h_{n,l}$$

Norming constants

We want to find

$$\int_{\mathbb{C}} P_k(x)\overline{P}_m(x)w(x)\mathrm{d}^2z = \langle P_k, P_m\rangle_w = \begin{cases} h_{n,m} & \text{if } m = k, \\ 0 & \text{if } m \neq k. \end{cases}$$

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$$\left\langle z^m, z^j \right\rangle_w = \int_{\mathbb{C}} z^m \bar{z}^j \, \mathrm{e}^{-nQ(z)} \, \mathrm{d}^2 z,$$

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Notice that

$$\begin{aligned} \langle z^m, z^j\rangle_w &= \int_{\mathbb{C}} z^m \bar{z}^j \, \mathrm{e}^{-nQ(z)} \, \mathrm{d}^2 z, \\ &= \int_0^\infty \int_0^{2\pi} r^{m+j} \, \mathrm{e}^{\mathrm{i}\theta(m-j)} \, \mathrm{e}^{-nq(r)} \, \frac{r}{\pi} \mathrm{d}r\mathrm{d}\theta, \\ &= \frac{1}{\pi} \int_0^{2\pi} \mathrm{e}^{\mathrm{i}\theta(m-j)} \, \mathrm{d}\theta \int_0^\infty r^{m+j+1} \, \mathrm{e}^{-nq(r)} \, \mathrm{d}r \end{aligned}$$

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&= \frac{1}{\pi} \int_0^{2\pi} e^{\mathrm{i}\theta(m-j)} \, \mathrm{d}\theta \int_0^\infty r^{m+j+1} \, e^{-nq(r)} \, \mathrm{d}r \\
&= \begin{cases} 0, & \text{if } m \neq j, \\ \int_0^\infty 2r^{2j+1} \, e^{-nq(r)} \, \mathrm{d}r, & \text{if } m = j. \end{cases}
\end{aligned}$$

Norming constants

Defining $z = r\,\mathrm{e}^{\mathrm{i}\theta}$ ($|z| = r$), because of the symmetry on Q we write

$$\begin{aligned} h_j &= \int_{\mathbb{C}} |z|^{2j}\, \mathrm{e}^{-nQ(z)}\, \mathrm{d}^2 z \\ &= \int_{\Theta}\int_{\mathbb{R}_+} r^{2j}\, \mathrm{e}^{-nq(r)}\, \frac{r}{\pi} \mathrm{d}r\mathrm{d}\theta, \\ &= 2\int_{\mathbb{R}_+} r^{2j+1}\, \mathrm{e}^{-nq(r)}\, \mathrm{d}r, \\ &= \int_{\mathbb{R}_+} 2r\, \mathrm{e}^{-n\left(q(r) - \frac{2j}{n}\ln(r)\right)}\, \mathrm{d}r. \end{aligned}$$

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We redefine our potential using

$$V_{\tau(j)}(r) = q(r) - 2\tau(j)\ln(r),$$

where $\tau(j) := j/n \in [0,1)$. Then, rewrite the integral as

$$h_j = \int_{\mathbb{R}_+} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r.$$

Taking a look at the first derivative of the newly defined potential we find that the critical point r_c must satisfy

$$V_\tau'(r) = q'(r) - \frac{2\tau}{r} \quad \Longrightarrow \quad r_c q'(r_c) = 2\tau.$$

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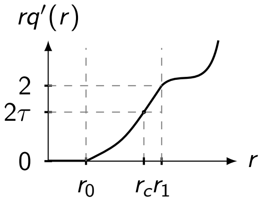
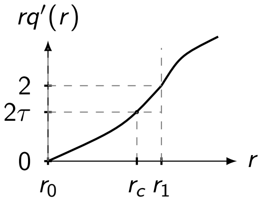
Denote $r_{\tau(j)}$ the r such that $rq'(r) = 2\tau(j)$. Also define the limit points when $\tau = 0$ and $\tau = 1$ as

$$\begin{cases} r_0, & \text{the largest solution of } rq'(r) = 0, \\ r_1, & \text{the smallest solution of } rq'(r) = 2. \end{cases}$$





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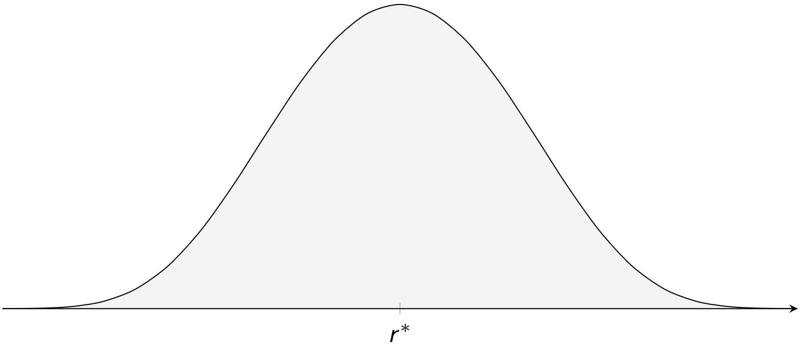

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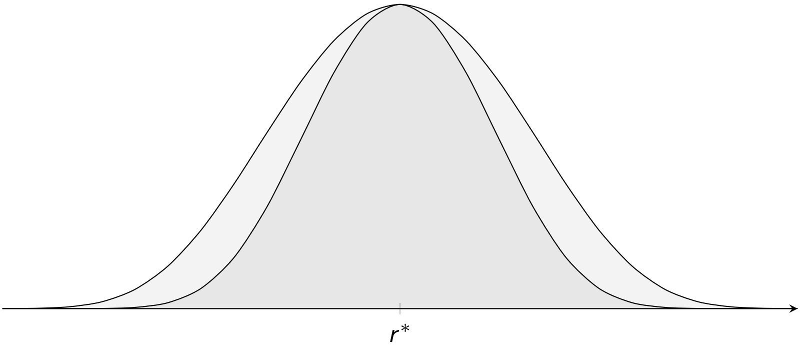

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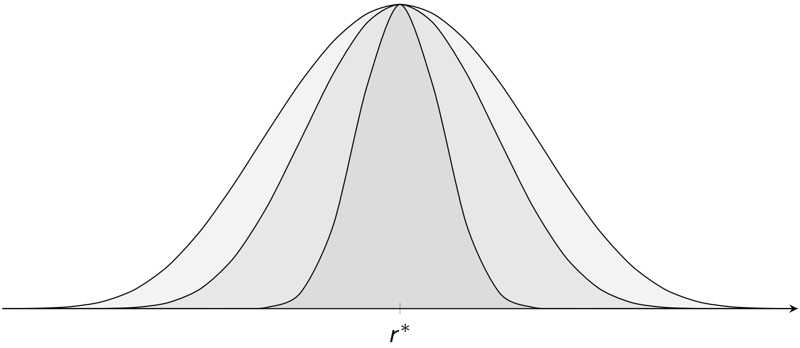

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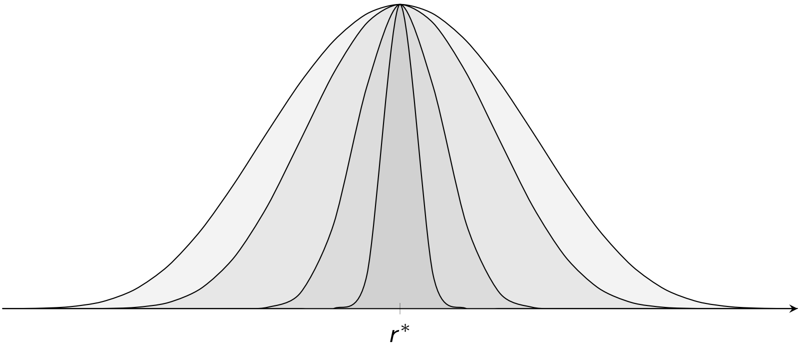

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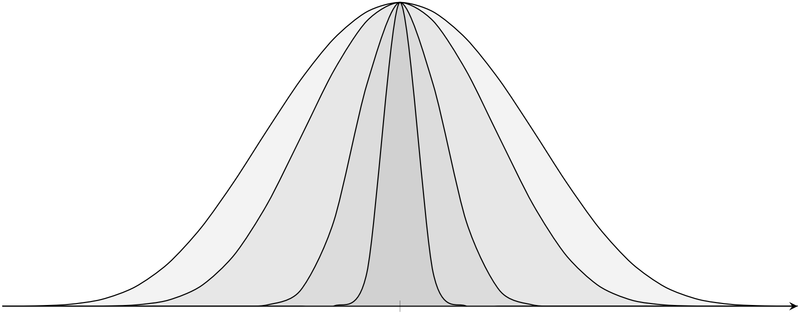

$$\lim_{n\to\infty} \frac{e^{-n(V_{\tau(j)}(r^*))}}{e^{-n(V_{\tau(j)}(r^*-\epsilon))}} = \lim_{n\to\infty} e^{-n(V_{\tau(j)}(r^*)-V_{\tau(j)}(r^*-\epsilon))} = \lim_{n\to\infty} e^{n|c|} \to +\infty.$$

Laplace Technique

We start by approximating $V_{\tau(j)}$ around its global minima by Taylor's theorem

$$V_{\tau(j)}(r) = V_{\tau(j)}(r^*) - \frac{1}{2}|V^{(2)}_{\tau(j)}(r^*)|(r - r^*)^2 + \epsilon((r - r^*)^3)$$

and then perform a change of variables $\frac{r}{\sqrt{n}} = s - r^*$, so that

$$\int_{r^* - \frac{\ln(n)}{\sqrt{n}}}^{r^* + \frac{\ln(n)}{\sqrt{n}}} 2s\, e^{-nV_{\tau(j)}(s)}\, ds \approx \frac{e^{-nV_{\tau(j)}(r^*)}}{\sqrt{n}} \int_{-\ln(n)}^{\ln(n)} 2\left(r^* + \frac{r}{\sqrt{n}}\right) e^{-\frac{1}{2}|V^{(2)}_{\tau(j)}(r^*)|r^2}\, dr$$

and

$$\int_{r^* + \frac{\ln(n)}{\sqrt{n}}}^{\infty} |2s|\, e^{-nV_{\tau(j)}(s)}\, d(s) \le e^{-nV_{\tau(j)}(r^*)}\, \epsilon_n$$

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and

$$\int_{r^*+\frac{\ln(n)}{\sqrt{n}}}^{\infty} |2s|\, \mathrm{e}^{-nV_{\tau(j)}(s)}\,\mathrm{d}(s) \leq \mathrm{e}^{-nV_{\tau(j)}(r^*)}\,\epsilon_n$$

Theorem

(Laplace's Method) Let Φ *be*

$$\Phi(n) = \int_a^b g(t)\, \mathrm{e}^{-n\phi(t)}\, \mathrm{d}t$$

where $g(t)\,\mathrm{e}^{-n\phi(t)}$ *is absolutely integrable over* $[a, b]$ *for every* n*, the function* $\phi(t)$ *attains its maximum only at the point* c *in* (a, b)*, the second derivative* $\phi''(t)$ *exists in a neighborhood of and is negative on* c*. Finally,* $g(t)$ *is non-zero and continuous at* c*. Then the function* Φ *satisfies an estimate of the form*

$$\Phi(n) = \mathrm{e}^{-n\phi(c)} \left(\frac{\sqrt{2\pi} g(c)}{\sqrt{|\phi''(c)|} n^{1/2}} + \mathcal{O}(n^{-3/2}) \right)$$

as $n \to \infty$*.*

Theorem

We have the following asymptotic of n for h_j, with $0 \leq j \leq n-1$,

$$h_j = \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{n}\mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2}) \right],$$

where we have set

$$\mathfrak{B}(r_{\tau(j)}) := -\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2}\frac{1}{8} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2}\frac{1}{2r_{\tau(j)}} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^3}\frac{5}{24}.$$

Quadrature

We have

$$\ln \mathcal{Z}_n^Q = \ln n! + \sum_{j=0}^{n-1} \ln h_j.$$

where

$$h_j = \int_{\mathbb{C}} |z|^{2j} \, \mathrm{e}^{-nQ(z)} \, \mathrm{d}^2 z \approx \frac{\mathrm{e}^{-nV_{\tau(j)}(r_c(\tau(j)))}}{\sqrt{n}}.$$

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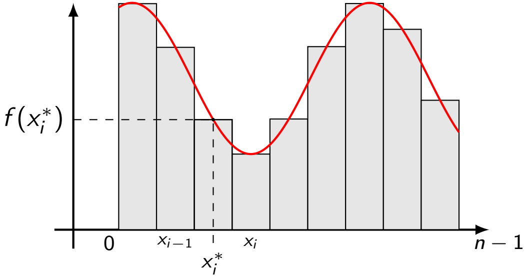
$$\Delta_n \sum_{j=0}^{n-1} \ln h_j \approx \int_a^b \ln h_j$$



Quadrature I

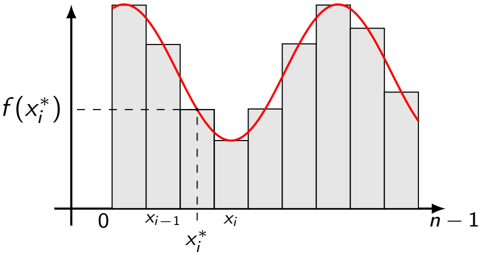
We begin with $h = (b-a)/n$, s.t.

$$\begin{aligned}
\int_a^b f(x)\mathrm{d}x &\approx h\sum_{j=0}^{n} f(x_j^*),\\
&= h\sum_{j=0}^{n} \frac{f(x_j)+f(x_{j-1})}{2},\\
&= h\left[\frac{f(a)}{2} + f(a+h) + \cdots + f(b-h) + \frac{f(b)}{2}\right],\\
&= h\sum_{j=1}^{n-1} f(x_j^*) + \frac{h}{2}\left(f(b)+f(a)\right),\\
&= h\sum_{j=0}^{n-1} f(x_j^*) + \frac{h}{2}\left(f(b)-f(a)\right).
\end{aligned}$$

Theorem

(Euler-Maclaurin Formula) If m and n are natural numbers and $f(x)$ is a real or complex valued continuous function for real numbers x in the interval $[m, n]$, then the integral can be approximated by the sum (or vice versa)

$$\sum_{i=m+1}^{n} f(i) = \int_m^n f(x)\mathrm{d}x + \frac{f(n) - f(m)}{2} + \sum_{k=1}^{\lfloor \frac{p}{2} \rfloor} \frac{B_{2k}}{(2k)!} \left(f^{(2k-1)}(n) - f^{(2k-1)}(m) \right) + R_p$$



We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$= \sum_{j=0}^{n-1} \ln \left[\frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2}) \right] \right],$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$= \sum_{j=0}^{n-1} \left[-nV_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2} \left(\ln\left(2\pi r_{\tau(j)}^2\right) - \ln n - \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) \right) + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2}) \right]$$

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Theorem

$$-n\sum_{j=0}^{n-1} V_{\tau(j)}(r_{\tau(j)}) = -n^2 I_Q[\mu_Q] + nU_{\mu_Q}(0) - \frac{1}{6}\ln\left(\frac{r_0}{r_1}\right) + \mathcal{O}(n^{-2}),$$

where

$$I_Q[\mu_Q] := \left[\frac{1}{4}\int_{r_0}^{r_1} q(s)sV^{(2)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1) \right],$$

$$-2U_{\mu_Q}(0) := \int_{r_0}^{r_1} \ln(s)(sq'(s))'\mathrm{d}s = q(r_1) - q(r_0) + 2\ln(r_1).$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$= -n^2 I_Q[\mu_Q] + nU_{\mu_Q}(0) - \frac{1}{6}\ln\left(\frac{r_0}{r_1}\right) + \sum_{j=0}^{n-1}\left[\frac{1}{2}\left(\ln\left(2\pi r_{\tau(j)}^2\right) - \ln n - \ln\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right)\right) + \frac{1}{n}\mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2})\right]$$

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Theorem

Asymptotically on n,

$$\sum_{j=0}^{n-1} \ln V_{\tau(j)}^{(2)}(r_{\tau(j)}) = nE_Q[\mu_Q] - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + \mathcal{O}(n^{-1}),$$

$$\sum_{j=0}^{n-1} \ln\left(2\pi r_{\tau(j)}^2\right) = n\ln(2\pi) + 2\left(-nU_{\mu_Q}(0) - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right) + \mathcal{O}(n^{-1})\right),$$

$$\frac{1}{n}\sum_{j=0}^{n-1} \mathfrak{B}(r_{\tau(j)}) = \frac{1}{4}\int_0^{r_1} \mathfrak{B}(s)sV^{(2)}(s)\mathrm{d}s + \mathcal{O}(n^{-2})$$

where

$$E_Q[\mu_Q] := \frac{1}{2}\int_{r_0}^{r_1} \ln\left(\frac{V^{(2)}(s)}{4}\right)(sq'(s))'\mathrm{d}s.$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$\sum_{j=0}^{n-1} \ln h_j = -n^2 I_Q[\mu_Q] + nU_{\mu_Q}(0) - \frac{1}{6}\ln\left(\frac{r_0}{r_1}\right) - \frac{n}{2}\ln(n) - \frac{1}{2}\left(nE_Q[\mu_Q] - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right)\right)$$
$$+ \frac{1}{2}\left(n\ln(2\pi) + 2\left(-nU_{\mu_Q}(0) - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right)\right)\right) + \frac{1}{4}\int_0^{r_1} \mathfrak{B}(s)sV^{(2)}(s)\mathrm{d}s + \mathcal{O}(n^{-1})$$

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$$\sum_{j=0}^{n-1} \ln h_j = -n^2 I_Q[\mu_Q] - \frac{n}{2}\ln(n) + n\left(\frac{\ln(2\pi)}{2} - \frac{1}{2}E_Q[\mu_Q]\right) - \frac{2}{3}\ln\left(\frac{r_0}{r_1}\right) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right)$$
$$- \frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\,\mathrm{d}r + \mathcal{O}(n^{-1}).$$

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$$- \frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\,\mathrm{d}r + \mathcal{O}(n^{-1}).$$

Theorem

Asymptotically on n,

$$\sum_{j=0}^{n-1} \ln h_j = -n^2 I_Q[\mu_Q] - \frac{n}{2}\ln n + n\left(\frac{\ln(2\pi)}{2} - \frac{1}{2}E_Q[\mu_Q]\right) + F_V[A_{[r_0,r_1]}] + \mathcal{O}(n^{-1})$$

where

$$F_V[A_{[r_0,r_1]}] := -\frac{2}{3}\ln\left(\frac{r_1}{r_0}\right) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\,\mathrm{d}r,$$

Theorem

For the annulus case, where $r_0 > 0$, we have the following expression for the free energy asymptotically on n given by

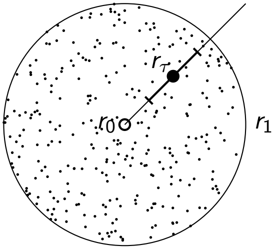
$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{n}{2}\left(\ln(2\pi) - E_Q[\mu_Q] - 2\right) + \frac{n}{2}\ln(n) + \frac{\ln(2\pi)}{2} + \frac{1}{2}\ln(n) + F_V[A_{[r_0,r_1]}] + \mathcal{O}(n^{-1})$$

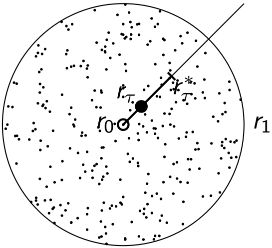
where

$$I_Q[\mu_Q] := \left[\frac{1}{4}\int_{r_0}^{r_1} q(s)sV^{(2)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1)\right],$$

$$E_Q[\mu_Q] := \frac{1}{2}\int_{r_0}^{r_1}\ln\left(\frac{V^{(2)}(s)}{4}\right)(sq'(s))'\mathrm{d}s.$$

$$F_V[A_{[r_0,r_1]}] := -\frac{2}{3}\ln\left(\frac{r_1}{r_0}\right) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2}r\mathrm{d}r,$$


r_τ
r_0
r_1


r_τ
r_τ^*
r_0
r_1

Theorem

For the disk case, where $r_0 = 0$, we have the following, asymptotically on n, for the free energy

$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{1}{2} n \ln n + \frac{n}{2}\left(\ln(2\pi) - 2 - E_Q[\mu_Q]\right) - \frac{5}{12} \ln n \frac{\ln(2\pi)}{2} + \zeta'(-1) + F_V[D_{r_1}] + \mathcal{O}(n^{-1/12}(\ln n)^3).$$

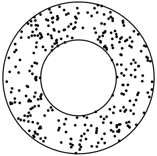
where

$$I_Q[\mu_Q] := \left[\frac{1}{4}\int_{r_0}^{r_1} q(s) s V^{(2)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2} q(r_1)\right],$$

$$E_Q[\mu_Q] := \frac{1}{2}\int_{r_0}^{r_1} \ln\left(\frac{V^{(2)}(s)}{4}\right)(sq'(s))'\mathrm{d}s,$$

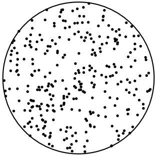
$$F_V[D_{r_1}] := \frac{1}{6}\ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{3}\ln(r_1) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(0)}\right) - \frac{1}{32}\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{1}{12}\int_0^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r.$$

For the annulus,



$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{1}{2} n \ln n + \frac{n}{2}\left(\ln(2\pi) - 2 - E_Q[\mu_Q]\right) + \frac{1}{2}\ln(n) + \frac{\ln(2\pi)}{2} + F_V[A_{[r_0,r_1]}] + \mathcal{O}(n^{-1}).$$

$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{1}{2} n \ln n + \frac{n}{2}\left(\ln(2\pi) - 2 - E_Q[\mu_Q]\right) - \frac{5}{12}\ln(n) + \frac{\ln(2\pi)}{2} + \zeta'(-1) + F_V[D_{r_1}] + \mathcal{O}(n^{-1/12}(\ln n)^3).$$



for the disk

Obrigado!

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_(____nm________ /____\___|'''''''''''|João V. A. Pimenta|'''''''''|
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